

Capacitor-less Stochastic Leaky-FeFET Neuron of Both Excitatory and Inhibitory Connections for SNN with Reduced Hardware Cost

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Abstract—For the unsupervised learning of spiking neural networks (SNNs), both excitatory and inhibitory signals with different arriving time should be connected to the artificial neurons, causing the high-hardware-cost issue for the implementation. In this work, a capacitor-less neuron based on the novel Leaky-FeFET (L-FeFET) device is proposed and experimentally demonstrated with remarkably reduced hardware cost of only two transistors and one resistor. For the first time, by exploiting the physics of directional ferroelectric polarization switching, both excitatory and inhibitory input connections can be emulated for the biomimetic neuronal dynamics in both experiments and simulations. Moreover, the neuronal stochastic behavior can be realized without additional hardware cost due to the inherent ferroelectric dynamics. Based on the proposed neuron, a bio-plausible SNN is efficiently implemented for clustering and inference tasks. Both biomimetic clustering by self-organizing map learning and high-accuracy (92%) inference by winner-take-all learning are realized. This work provides a promising highly-integrated neuron solution for brain-inspired neuromorphic computing system.

I. INTRODUCTION

The implementations of basic computing elements based on novel devices for brain-inspired neuromorphic computing have triggered a lot of interests. We have recently proposed a novel leaky ferroelectric FET (L-FeFET) device and experimentally demonstrated its superiority for the artificial excitatory neurons realization [1]. In addition to the excitatory inputs, the inhibitory inputs, which usually exhibit different arriving time, are also indispensable for the homeostasis of biological neurons and unsupervised learning capacity of spiking neural networks (SNNs) (Fig. 1). However, most of the current neuron schemes which can emulate both excitatory and inhibitory connections are very complex and implemented with several large capacitors, suffering from high hardware cost [2-3]. Recently a capacitor-less neuron based on phase-change devices for both excitatory and inhibitory behavior treatment was demonstrated [4], while it still requires at least two resistors and ten transistors for input signal processing (Fig. 2b), along with additional differential circuit for excitatory-inhibitory signals comparing.

In this work, a novel capacitor-less neuron based on L-FeFET is proposed and experimentally demonstrated enabling both excitatory and inhibitory input connections by using only one resistor and two transistors, while simultaneously guaranteeing the neuronal stochastic behavior. The unsupervised SNN based on our proposed neurons is also implemented, realizing both the biomimetic clustering through

self-organizing map learning and the high-accuracy inference by winner-take-all learning with greatly-saved hardware cost.

II. L-FEFET NEURON DESIGN WITH EXCITATORY AND INHIBITORY INPUT CONNECTIONS

A. Capacitor-less neuron design for biomimetic dynamics

The novel neuron circuit topology without capacitor is designed based on our proposed L-FeFET device in [1], as shown in Fig. 2a, which only consists of a resistor, a conventional nFET for excitatory and inhibitory input conversion, and a L-FeFET for neuron dynamics.

Excitatory and Inhibitory input accumulation: When the voltage signals of excitatory and inhibitory inputs are applied respectively, the conversion circuit of the proposed design generates intermediate signals, which will directly be applied to the gate of L-FeFET. The excitatory input is applied through a low resistance R_0 , while the inhibitory input can turn on the pull-down transistor of nFET, and then the negative voltage V_{inh} can be applied to the gate of L-FeFET. The value of V_{inh} should be designed to ensure reverse polarization switching of FE layer in L-FeFET. According to the nucleation and growth of FE domains theory, progressive forward or reverse polarization switching depend on both external electric field and time [5]. Therefore, the polarization in L-FeFET can be switched and accumulated with applied voltage spikes, and no capacitors are needed. Since the threshold voltage (V_{TH}) and channel conductivity of L-FeFET can be modulated by FE polarization charge, the neuron circuit with only 2T1R shows the accumulation effect with excitatory and inhibitory inputs, mimicking the excitatory and inhibitory stimuli integration of neurons by ultra-low hardware cost.

Leaky effect: Conventional FeFET designed for non-volatile memory or synapse application requires long retention time, which cannot realize the leaky effect of neuron. In L-FeFET, both the depolarization field and the leaky current of FE layer are enlarged to realize the accelerated FE polarization degradation, emulating the leaky effect of neuronal membrane potential.

Stochastic behavior: The neuronal stochastic behavior can be realized without extra hardware, due to the random dynamic process of ferroelectric domains nucleation [6]. Stochastic behaviors of neurons benefit information processing, such as enabling SNNs to carry out probabilistic inference through sampling [7].

B. Neuron behavior based on L-FeFET modeling

To demonstrate the behavior of designed neuron topology, the L-FeFET device model is established based on dynamic Preisach ferroelectric switching model [8] and BSIM-4 MOSFET model. The leaky effect in L-FeFET by equivalently increasing the depolarization field in FE layer can be reflected in the model (Fig. 3). HSPICE circuit simulation based on L-

FeFET model shows that the neuron design can emulate the leaky-integrate-and-fire (LIF) function with both excitatory and inhibitory inputs (Fig. 4a&b). However, after firing, for the conventional FeFET based neuron circuit, the V_{TH} of FeFET will maintain low due to the nonvolatile characteristics, and thus the neuron cannot further perform normal neuronal dynamics without additional erase circuit, which indicates the necessity of leaky effect in FeFET for low-hardware-cost neuron design.

III. EXPERIMENTAL DEMONSTRATION

A. L-FeFET characteristics for excitement and inhibition

In this experiment of L-FeFET, the leaky FE layer is 5nm $HfO_5ZrO_5O_2$ film with special treatment (details are presented in our previous work [1]). Fig. 5 shows I_d - V_g hysteresis of L-FeFET and measuring method. With positive spikes applied to the gate of L-FeFET, the FE polarization will switch (Fig. 6a), and thus the device drain current (I_d) continuously increases. When applying negative spike train after sufficient positive spikes, the reverse polarization switching can induce the rapidly decreased I_d (Fig. 6b). It shows that the FE polarization switching strongly depends on the voltage and width of the positive and negative pulses. Those characteristics demonstrate the capability of L-FeFET to emulate neuronal dynamics of LIF with excitatory and inhibitory inputs.

B. L-FeFET neuron with excitatory & inhibitory input connections

Based on the L-FeFET, the proposed neuron circuit topology is experimentally constructed with 2R1T as shown in Fig. 7. Biologically similar excitatory and inhibitory stimuli are applied to the corresponding ports by pulse generator units, and the neuron output is obtained through monitoring the I_d .

Basic behavior with excitatory and inhibitory inputs: As shown in Fig. 8-10, the L-FeFET neuron can implement the basic LIF function of neurons with excitatory input spikes applied. With excitatory input spikes applied, V_{TH} gradually decrease due to polarization switching of FE layer. When the V_{TH} is low enough, the neuron generates an output spike (Fig.8). The leaky property of biological neuron can also be mimicked as shown in Fig. 9. Due to the FE polarization degradation, the neuron can be reset to the initial state after firing (Fig. 10). When inhibitory stimuli are applied in the interval of excitatory stimuli trains, the spike firing behavior of the neuron will be mitigated (Fig. 11b), compared with the output without inhibitory stimuli in Fig.11a. Since inhibitory stimuli cause reverse polarization against excitatory stimuli, the V_{TH} of L-FeFET increases and I_d decreases, which emulate decreased membrane potential of biological neurons with inhibitory postsynaptic potential. So the neuron requires more excitatory stimuli for next firing. When the inhibitory stimuli with the larger $|V_{inh}|$ (Fig. 11c) or the longer pulse width (Fig. 11d) are applied to the neuron, excessive reverse polarization will accumulate, causing dramatic V_{TH} increase. the same number of excitatory stimuli cannot make L-FeFET sufficiently conductive, thus the neuron cannot generate spike output.

Stochastic behavior: From the initial state, the L-FeFET based neuron requires several excitatory pulses to reach the threshold of activation and fire. In our experiment, the number of accumulated excitatory pulses required is not fixed for every measurement, and shows a Gaussian distribution (Fig. 12) due to the stochastic dynamic process of ferroelectric domains nucleation in L-FeFET.

The homeostasis of excitation and inhibition: When a neuron receives strong continuous excitatory stimuli without any inhibitory input, it may fall into an abnormal fire pattern. However, with suitable inhibitory stimuli applied to the neuron, its spike firing behavior can be in a dynamically balanced state (Fig. 13). That behavior is similar to the biological observations that when inhibition is blocked, the cortical activity may become epileptic and the neurons may lose selectivity to different stimulus features. The cortical neurons can only normally function when receiving balanced excitatory and inhibitory synaptic inputs [9].

IV. DEMONSTRATION OF UNSUPERVISED SNN

Based on the proposed stochastic L-FeFET neuron with both excitatory and inhibitory input connections, the unsupervised SNN consisting of two-layer neurons (Fig. 14) are simulated for the realizations of the biomimetic clustering through self-organizing map learning and the high-accuracy inference by winner-take-all learning.

For self-organizing map learning [10], the connection topology of the output layer (OL) includes both local excitatory and lateral inhibitory connections (Fig. 15a). Therefore, once a neuron in OL fires, it can induce the nearby neurons to fire with the same inputs and learns the similar filters. The unsupervised learning results in Fig. 15c demonstrate the biomimetic clustering behavior, emulating the topologically ordered spatial representations of features in the primate cortex [11].

For winner-take-all learning, the OL connection topology has only lateral inhibitory connections, which indicates that the winner neuron can inhibit all the other neurons in the OL. The learning results show that this network can achieve high-accuracy inference of 92%.

In addition, benefiting from the excitatory and inhibitory input connections of the novel L-FeFET neuron, the output neurons can transmit the excitatory and inhibitory signals in the layer without additional inhibitory layer neurons, saving almost half number of the neurons for SNN. The proposed neuron with only 2T1R provides a highly-integrated neuron solution for brain-inspired neuromorphic computing systems.

V. SUMMARY

A capacitor-less L-FeFET neuron enabling both excitatory and inhibitory input connections with two transistors and one resistor (Table.1) is proposed and experimentally demonstrated to emulate biological neuronal dynamics. The new neuron with dramatically-reduced hardware cost is applied to unsupervised SNN with the implementation of biomimetic clustering and high-accuracy (92%) inference, showing great potentials for highly-integrated neuromorphic computing system.

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References [1] C. Chen, et al., *VLSI*, pp. T136, 2019. [2] R. Wang, et al., *BioCAS*, pp. 1-4, 2015. [3] Z. Wang, et al., *IEDM*, pp. 13.3.1, 2018. [4] T. Tuma, et al., *Nature Nanotech.*, 11, pp. 693, 2016. [5] H. Orihara, et al., *J. Phys. Soc. Jpn.*, 63, 1031, 1994. [6] Y. H. Shin, et al., *Nature*, 449, pp. 881, 2007. [7] W. Maass, et al., *Proc. of IEEE*, 102.5, pp. 860, 2014. [8] K. Ni, et al., *VLSI*, pp. 131, 2018. [9] Rubin, et al., *PNAS*, 114.44, pp. E9366, 2017. [10] H. Hazan, et al., *IJCNN*, pp. 1-6, 2018. [10] M. Meister, et al., *J. Neurosci.* 38 (10) 2430-2441, 2018.

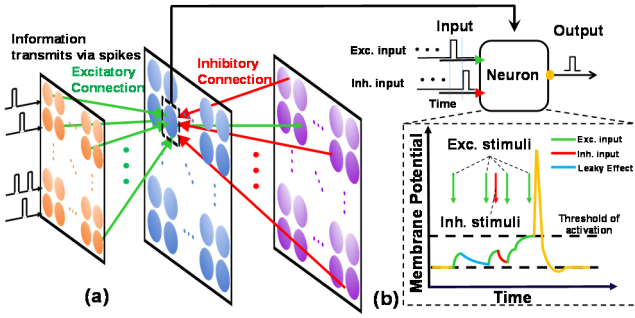


Fig.1 (a) The schematic representation of typical brain-inspired neuromorphic computing system: spiking neural network (SNN) consisting of excitatory connections, inhibitory connections and high density of artificial neurons. (b) The biomimetic dynamics of artificial neuron with excitatory (exc.) and inhibitory (inh.) inputs.

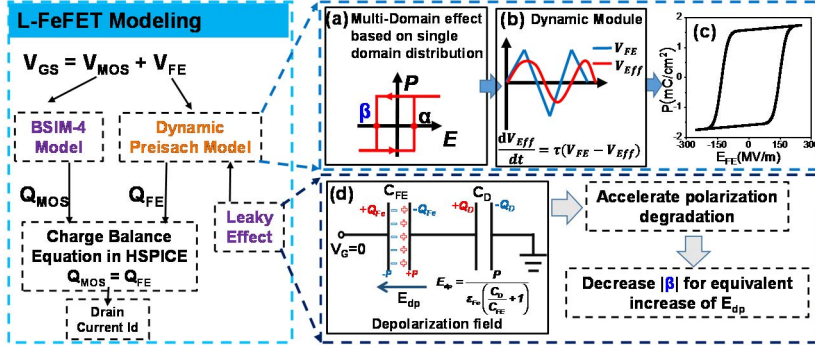


Fig.3 The modeling framework of the L-FeFET in this work. (a)(b) Dynamic Preisach model and (c) P-V loop of the FE model. (d) the depolarization field E_{dp} can be enlarged to accelerate the polarization degradation to achieve the leaky effect of ferroelectric layer.

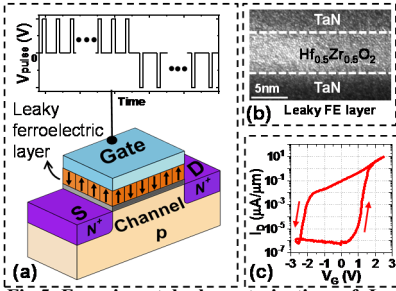


Fig.5 Experimental characterization of L-FeFET. (a) The applied gate pulses. (b) TEM image of FE layer. (c) L-FeFET structure. (d) Measured I_D - V_G curve of L-FeFET.

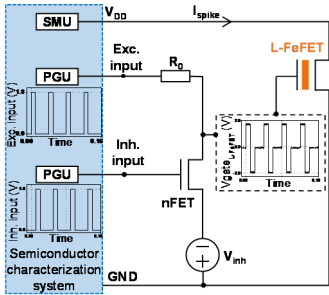


Fig.7 Measurement setup of L-FeFET neuron, where pulse generator units (PGU) generate exc. and inh. inputs; source measure unit (SMU) monitors I_{spike} .

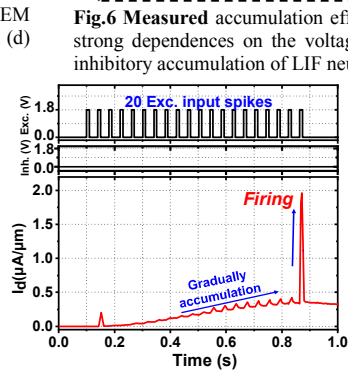


Fig.8 Measured integrate-and-firing behavior of L-FeFET neuron. It can emulate the gradual accumulation and rapid spike generation of biological neuron.

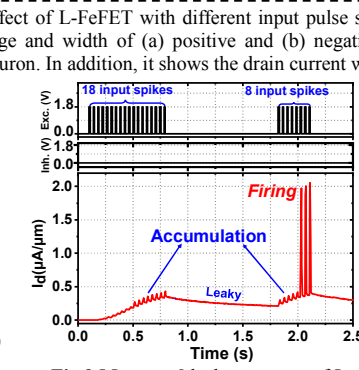


Fig.9 Measured leaky property of L-FeFET neuron. The FE polarization degrades during the free interval of input exc. spike trains.

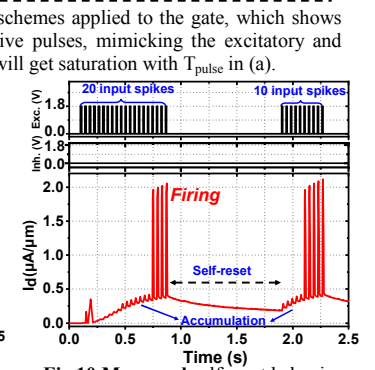


Fig.10 Measured self-reset behavior during continuous bursts due to FE polarization degradation. The neuron restarts LIF function after self-reset.

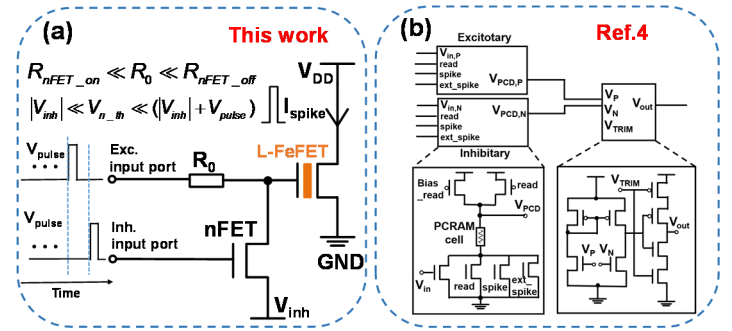


Fig.2 (a) The proposed L-FeFET neuron with excitatory and inhibitory input connections. R_{nFET_on} (R_{nFET_off}) is the on (off) channel resistance of the nFET. V_{pulse} is the input pulse voltage. V_{inh} is the inhibitory negative voltage. (b) The neuron based on phase-change devices [4].

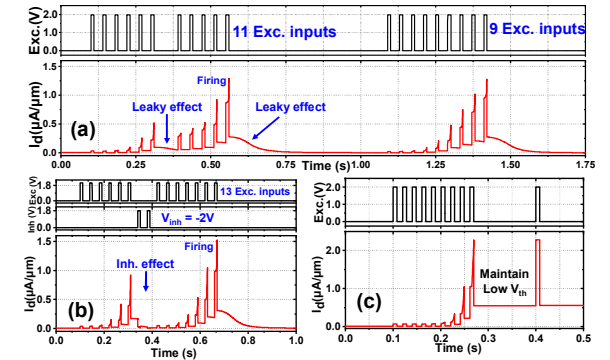


Fig.4 Simulated L-FeFET based neuron behavior: (a) LIF behavior. (b) exc. and inh. inputs accumulation; it takes more exc. pulses to fire with inhibition. (c) Without leaky effect, FeFET maintains low V_{th} after firing.

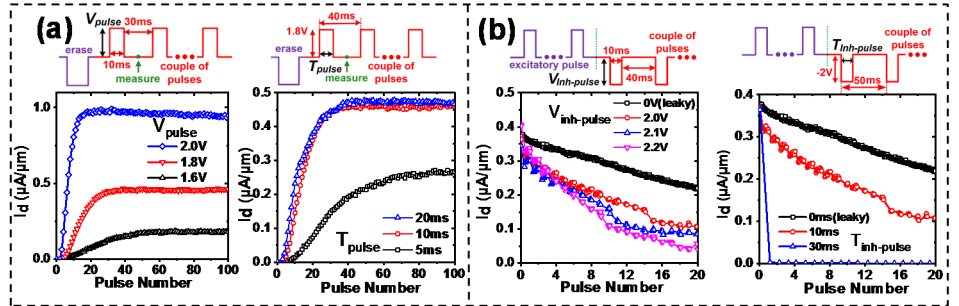


Fig.6 Measured accumulation effect of L-FeFET with different input pulse schemes applied to the gate, which shows strong dependences on the voltage and width of (a) positive and (b) negative pulses, mimicking the excitatory and inhibitory accumulation of LIF neuron. In addition, it shows the drain current will get saturation with T_{pulse} in (a).

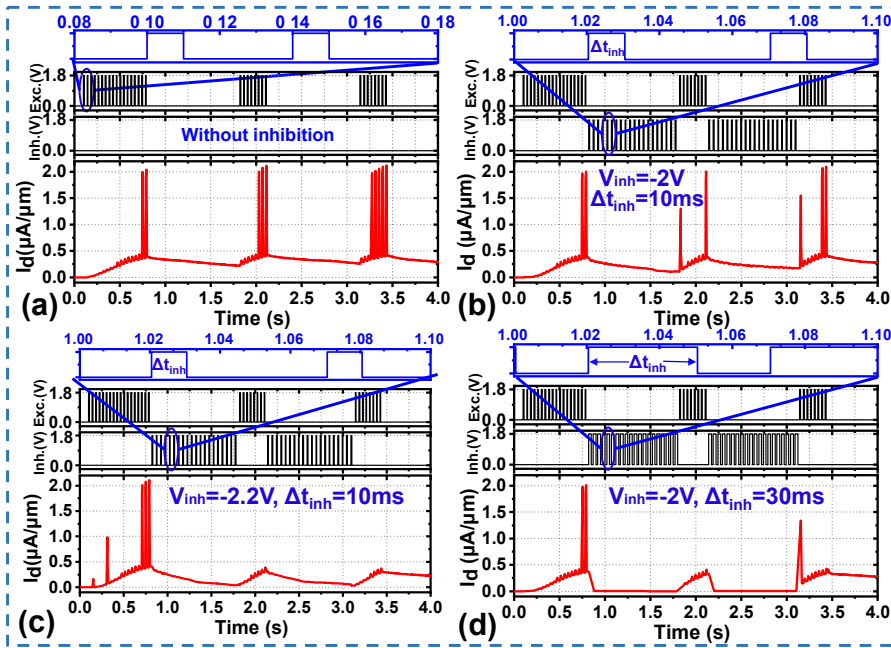


Fig.11 Measured spike firing behaviors with the same exc. inputs and the different inh. inputs. The period of exc. signal is 40ms and the pulse width is 10ms, and there is interval free time for inhibition. (a) Without inhibition, the neuron is bursting. (b) With inhibitory voltage V_{inh} of -2V, and inhibitory pulse width Δt_{inh} of 10ms in a inh. input period, the neuron firing is weakened and it generates sparse burst-spike trains. (c) With V_{inh} of -2.2V and Δt_{inh} of 10ms or (d) V_{inh} of -2.0V and Δt_{inh} of 30ms, the neuron firing is completely suppressed and cannot generate spike output, which indicates V_{inh} and Δt_{inh} can modulate the inhibitory intensity.

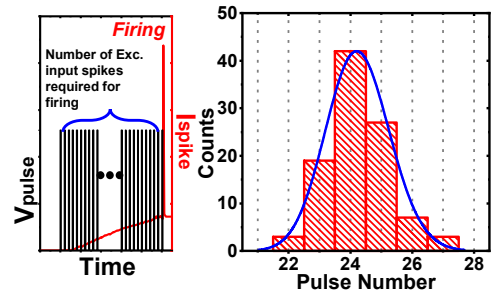


Fig.12. Measured stochastic behavior of L-FeFET neuron. Histogram of the number of input pulses required to fire the L-FeFET neuron showing with a Gaussian distribution.

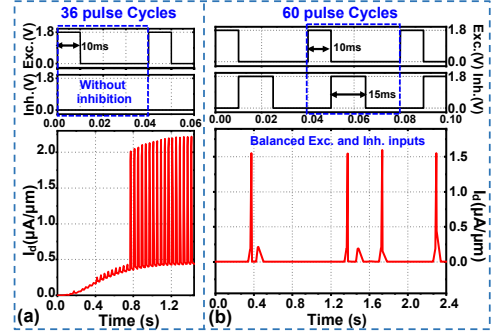


Fig.13 Measured spikes when the neuron receives strong continuous exc. stimuli (a) without inh. stimuli (may fire abnormally strong); and (b) with moderate inh. stimuli (a dynamically balanced state).

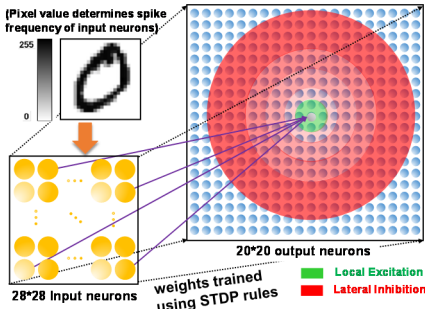


Fig.14 The architecture of SNN based on L-FeFET neurons with exc. and inh. input connections. The output layer (OL) may consist of local excitatory and lateral inhibitory connections.

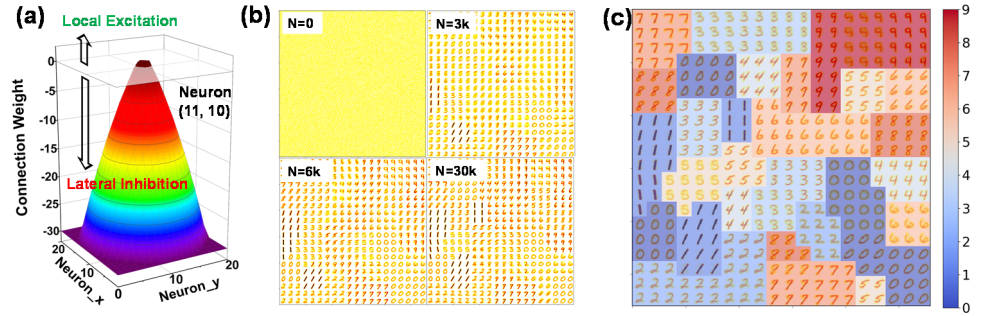


Fig.15 Self-organizing map learning: (a) The connection weight topology of a centered neuron to other neurons in output layer, which shows exc. connections with adjacent neurons and lateral inhibitory connections whose intensity increases with the distance between neurons. (b) Learning process over training epochs (N) of 0, 3000, 6000, and 30000. (c) The biomimetic clustering emulates the topologically ordered spatial representations of features in the primate cortex

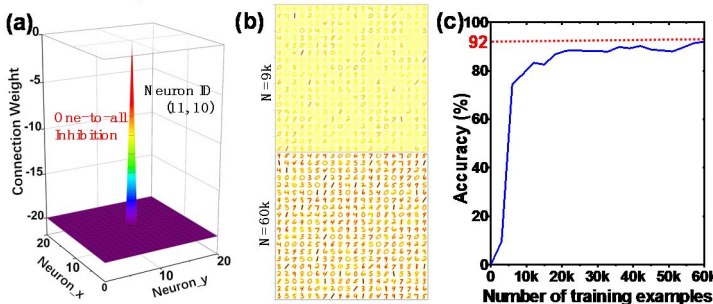


Fig.16 Winner-take-all learning: (a) The strong lateral inhibitory connections between OL neurons. Once a neuron in OL fire, it inhibits all the other neurons in OL. (b) The learning process over train epoch (N) of 9k and 60k (c) The accuracy develops during training, and can get 92%, which means high-accuracy inference of SNN.

Table.1 Benchmark of bio-inspired neurons with excitatory and inhibitory input connections in this work with other reported solutions.

	Ref. [2]	Ref. [3]	Ref. [4]	This work
Device type	CMOS	Ferroelectric FET	PCRAM	Leaky-FeFET
w/ or w/o capacitor				
Physics	Capacitor charging	Abrupt hysteretic	Phase change	Polarization switching
Hardware cost	2C+16T	1FeFET+1T + Leaky-integrator circuit (2C+4R)	2R+10T w/ peripheral circuit	1L-FeFET + 1T1R
Stochastic behavior	/	/	Yes	Yes
Operation Voltage	1.2V	3.3V	6V	1.8V
Functionality of SNN	/	Unsupervised clustering	Signal representation	Unsupervised biomimetic clustering & High-accuracy inference